# ICD-11 CODER — READINESS REVIEW

Sugen assessment | concise team handout | reviewed repository commit eecbeb5

**NOT READY FOR SUBMISSION**

The offline JSONL pipeline works, but clinical ranking scores **7.1 / 75**. The main bottleneck is retrieval: the correct code is among the model candidates for only **20 of 60** development notes.

| 7.1 / 75 | 20 / 60 | 21 / 21 | 97 sec |
|---|---|---|---|
| dev score | gold code retrieved | tests passed | 600-note run* |

## WHAT WORKS

- **No-key mode** produces predictions without API credentials.
- **Output contract** held on the dev set: 60/60 IDs, no missing rows, no invalid codes, no duplicate IDs.
- **Incremental writes** flush after each prediction.
- **Automated suite:** all 21 tests pass.

## TOP TECHNICAL GAPS

- **Low accuracy:** top-1 4/60 (6.7%); top-3 9/60 (15%). Ambient-note top-1 is 0/15.
- **Retrieval ceiling:** 40/60 gold codes are absent from the top-20 candidates; model selection cannot recover them.
- **Clinical evidence:** negations are extracted but ignored; note sections merge; simple diagnoses receive unsupported detail (for example, “hypertension” becomes “ocular hypertension”).
- **Language coverage:** ASCII and English-title matching performs poorly on Spanish and non-Latin text.

## REQUIREMENT STATUS

- **MET in tested runs** — JSONL shape, ID preservation, valid codes, no-key fallback, incremental output.
- **OPEN** — exact Docker command, clean-clone build, target hardware limits, final bundle, deadline.

## OPTIONAL GEMMA PATH

- Compose sets `OLLAMA_HOST`, but the client ignores it and calls container loopback.
- A malformed response can crash the run; failures are retried for every note. At 20 seconds each, 600 timeouts could exceed the 60-minute cap.
- Reported Gemma output matches deterministic output; current logs do not prove when the model actually contributed.

## SUBMISSION DOCUMENTS

- **README is a placeholder**; required score, hidden-score estimate, data sources, extra evaluation, AI-tooling error, and hours are missing.
- **DECISIONS.md is incomplete:** 3 entries, all “commit pending”; no final bundle or numeric hours statement found.
- No completed evaluation beyond the 60 dev notes is demonstrated.

**PRIORITY** 1. Improve candidate retrieval and evidence handling; measure top-1/top-3 after each change. 2. Verify the exact Docker command from a clean clone and benchmark on grader-equivalent hardware. 3. Complete the README, decision log, independent evaluation, hours statement, and bundle.

*The 600-note run repeated the 60 development notes 10 times with new IDs and no gold labels. It measures throughput only, on this host—not hidden-set accuracy or target-machine performance. Docker was unavailable for this review. Source: supplied BRIEF.md and score.py; app ranking, evidence, retrieval, Ollama client; repository analysis/gap-analysis.md.